

GSM Architecture

Module 39

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0:02 [Music]

0:17 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:26 we will study about GSM system architecture first of all please see the contents which I shall

0:33 cover after covering brief introduction about GSM I shall take up GSM system

0:39 architecture in last I shall take up GSM services so now let us start global

0:46 system for mobile communication that is gsm is a globally accepted standard for digital cellular

0:54 communication digital seller networks are the segment of the market for mobile and wireless devices which are growing

1:02 day to-day they are the wireless extensions of traditional pstn or ISDN networks and allow for seamless roaming

1:11 with the same mobile phone Nation or even worldwide today these systems are mainly used for voice traffic however

1:20 data traffic is continuously growing and there are several Technologies for wireless data transmission using

1:27 Cellular Systems the most popular digital system is gsm with approximately 70% market share

1:36 GSM is the most successful digital mobile telecommunication system in the world it is used by over 800 million

1:45 people in more than 190 countries in the early 1980s Europe had numerous

1:51 coexisting analog mobile phone systems which were often based on similar standards but ran on slightly different

2:00 carrier frequencies to avoid the situation for a second generation fully digital system the group special Mobile

2:08 GSM was founded in 1982 this system was soon named the global system for mobile Communications

2:17 GSM with the specification process lying in the hands of European telecommunications Standard Institute

2:26 etsi in the context of universal mobile telecommunication system UMTS and the creation of

2:35 3gpp third generation partnership project the whole development process of GSM was transferred to

2:42 3gpp and further development is combined with 3G development 3G PP assigned new

2:49 numbers to all GSM standards the primary goal of GSM was to provide a mobile phone system that allows users to roam

2:58 throughout Europe and provides voice Services compatible to ISDN and other

3:04 pstn systems GSM is a typical second generation system replacing the first generation analog systems but not

3:13 offering the high worldwide data rates that the third generation systems such as UMTS GSM has initially been deployed in

3:22 Europe using 890 to 915 MHz for uplinks and 935 to 960 MHz for down LS now the

3:32 system is also called gsm900 to distinguish it from the later versions these versions comprise GSM at

3:42 1800 MHz which is called DCs that is digital cellular system 1800 and the GSM

3:50 system mainly used in us at 1900 mhz which is called PCS that is personal

3:57 Communications service 9 1900 GSM system architecture GSM comes with a hierarchal

4:05 complex system architecture comprising many entities and interfaces a GSM system consists of three subsystems

4:15 radio subsystem RSS Network and switching subsystem NSS and operation subsystem that is

4:24 OSS each subsystem will be discussed in more detail in the following sections generally a GSM customer only notices a

4:33 very small fraction of the whole network the mobile stations Ms and some antenna of Base transceiver stations that is

4:43 BTS Radio subsystem the radio subsystem RSS comprises all radio specific entities

4:52 that is the mobile stations and the base station subsystem GSM architecture pair in the

4:59 figure shows the connection between the RSS and the NSS via the a interface that

5:07 is solid lines and the connection to the OSS via the O interface these are the dash lines the a interface is typically

5:16 based on circuit switched PCM 30 systems carrying up to 3064 kilobits per second connections

5:26 whereas the O interface uses the signaling system number seven based on x.25
carrying management data to and
5:35 from the radio subsystem base station subsystem that is BSS a GSM network
comprises many BSS
5:46 each controlled by a base station controller the BSS performs all functions
necessary to maintain radio
5:54 connections to an MS coding decoding of voice and rate adaptation to and from
6:01 6 minutes, 1 second the wireless network part the base station subsystem contains
several base transceiver
6:10 stations base transceiver station BTS a BTS comprises all radio equipment
6:18 that is antennas signal processing amplifiers necessary for radio transmission a
BTS can form a radio cell
6:27 or using sectorized Ina and several cells are connected to Mobile station via the um
interface and to the BSC
6:37 while the abis interface the um interface contains all the mechanisms necessary for
wireless
6:46 transmission that is tdma fdma ETC a GSM cell can measor between some 100 m
and
6:55 35 kilomet depending on the environment base station controller that is BSC the
7:04 BSC basically manages the BTS it reserves radio frequencies
7:10 handles the Handover from one BTS to another within the BSS and performs
7:17 paging of the MS the BSC also multiplexes the radio channels onto the
7:23 fixed network connections at the a interface Mobile station that is Ms
7:30 mobile stations are most widely known cell or mobile phones are the section of a
GSM cellular network that the user
7:39 sees and operates in recent years their size has fallen dramatically while the level
of functionality has greatly
7:48 increased a further Advantage is that the time between charges has significantly
increased there are number
7:56 of elements to the cell phone although the two main elements are the main
hardware and the Sim the hardware itself contains the
8:05 main elements of the mobile phone including the display case battery and the
electronics used to generate the
8:13 signal and process the data receiver and to be transmitted it also contains a
number
8:20 known as the international mobile equipment identity that is IMI this is installed in
the phone at
8:29 Manu picture and cannot be changed it is accessed by the network during
registration to check whether the

8:36 equipment has been reported as stolen the Sim or subscriber identity module contains the information that

8:46 provides the identity of the user to the network it contains a variety of information including a number known as

8:54 the international mobile subscriber identity that is imsi this SIM card contains many identifiers

9:02 and tables such as card type serial number a list of subscribed Services a personal identity number pin a pin

9:12 unblocking key p and authentication key Ki and the international mobile

9:19 subscriber identity that is imsi the PIN is used to unlock the MS

9:26 using the wrong PIN three times will lock the Sim in such cases the p is

9:33 needed to unlock the Sim the MS stores Dynamic information while logged onto the GSM system such as example the

9:42 cipher key KC and the location information consisting of a temporary mobile subscriber identity that is tmsi

9:51 and delocation area identification that is L AI typical Ms for GSM 9 00 have a

10:00 transmit power of up to 2 wat whereas for GSM 1800 1 wat is enough due to the

10:09 smaller Sal size apart from the telephone interface an MS can also offer other types of

10:16 interfaces to users with display loudspeaker microphone and programmable soft keys further interfaces comprise

10:25 computer modems or Bluetooth typical Ms example mobile phones

10:33 comprise many more vendor specific functions and components such as cameras fingerprint sensors calendars address

10:42 books games and internet browsers personal digital assistance PDA with mobile phone functions are also

10:51 available the listeners should be aware that an MS could also be integrated into a car or be used for location tracking

10:59 of a container Network and switching subsystem the heart of the GSM system is

11:06 formed by the network and switching subsystem that is NSS the NSS connects the wireless

11:13 network with standard public networks and performs handovers between different base station subsystems it comprises

11:22 functions for worldwide localization of users and supports charging accounting and roaming of users between different

11:30 providers in different countries the NSS consist of the following switches and databases mobile switching center that

11:39 is MSC the main element within the core Network area of the overall gsm network architecture is the mobile switching

11:48 center that is MSC the MSC acts like a normal switching node within a pstn or ISDN but also

11:57 provides additional functionality to enable the requirements of a mobile user to be supported they set up connections

12:05 to other msc's to and to bsc's Via the a interface and form the fixed backbone

12:13 network of a GSM system an MSE manages several bsc's in a geographical region

12:21 these include registration authentication call location inter MSC handovers and call routing to a mobile

12:31 subscriber it also provides an interface to pstn so that calls can be rooted from the mobile network to a phone connected

12:39 to a landline interfaces to other ms's are provided to enable calls to be made to Mobiles on different networks msse

12:49 can also connect to public data networks PDN such as x.25 an MC handles all

12:57 signaling needed for connection contion setup connection release and Handover of connections to other

13:04 ms's home location register hlr the hlr is the most important

13:11 database in a GSM system it stores all user relevant information this comprises

13:19 static information such as the mobile subscriber ISDN number that is msisdn subscribe Services example call

13:28 forwarding roaming restrictions GPRS and the international mobile subscriber identity that is imsi damic information

13:37 is also needed example the current location area of the MS the mobile

13:44 subscriber roaming number that is msrn the current vlr and MSC as soon as an MS leaves its current

13:53 LA the information in the hlr is updated when a user switches on their phone the

14:00 phone registers with the network and from this it is possible to determine which BTS it communicates with so that

14:08 incoming calls can be rooted appropriately even when the phone is not active but switched on it re-registers

14:17 periodically to ensure that the network hlr is aware of its latest position this information is necessary

14:26 to localize a user in the world while G M Network all these user specific information elements only exist once for

14:35 each user in a single hlr which also supports charging and

14:41 accounting hlrs can manage data for several million customers and contain

14:48 highly specialized databases which must fulfill certain realtime requirements to answer requests within certain time

14:56 bounds there is one HL per Network although it may be distributed across

15:03 various subcenters to for operational reasons visitor location register that

15:11 is VR the vlr associated to each MSC is the dynamic database which stores all
15:19 information needed for the MS users currently in the location area that is
associated to the
15:27 MSC example I MSI msisd n hlr address if a new Mobile station comes into a
15:36 location area La the VR is responsible for this process and it copies all
15:43 relevant information for this user from home location register this hierarchy of
15:49 vlr and hlr avoids frequent hlr updates and long distance signaling of user
15:57 information the vlr can be implemented as a separate entity but it is commonly
realized as a integral part of the msse
16:06 rather than a separate entity in this way exess is made faster and more convenient
equipment identity register
16:16 e the E is the entity that decides whether given mobile equipment may be allowed
onto the network each mobile
16:25 equipment has a number known as the international mobile equipment identity this
number is installed in the
16:33 equipment and is checked by the network during registration authentication Center
Au
16:41 the Au is a protected database that contains the secret key also contained in the
users's SIM card it is used for
16:49 authentication and for ciphering on the radio channel Gateway mobile switching
Center
16:57 gmsc the GM SC is the point to which a mobile equipment terminating call is
initially rooted without any knowledge
17:06 of the MS location the gmsc is thus in charge of obtaining the msrn that is
17:14 Mobile station roaming number from the hlr based on the msisd n Mobile station
ISDN number the
17:23 directory number of a Mobile station and routing the call to correct visited
17:29 mobile switching service center the MSC part of the term
17:35 gmsc is misleading since the Gateway operation does not require any linking to an
17:43 MSC SMS gateway msg the msg or SMS gateway is the term that is used to
17:52 collectively describe the two short message Services gateways defined in the GSM
standards the two gateways handle
18:00 messages directed in different directions the SMS gmsc short message
18:07 service Gateway mobile switching Center is for short messages being sent to a
18:12 mobile equipment the SMS iwmc short message service interworking
18:21 mobile switching Center is used for short messages originated with the mobile on
that Network the SMS gmsc role
18:31 is similar to that of gmsc whereas the SMS iwmc provides a fixed access point to

18:40 the short message service center operation subsystem the third part of a GSM system the operational

18:50 subsystem that is OSS contains the necessary functions for Network operation and

18:57 maintenance the old SS processes its own network entities and accesses other

19:04 entities via ss7 signaling the operation support subsystem is an element within the

19:11 overall gsm network architecture that is connected to components of the NSS and the

19:17 BSC it is used to control and monitor the overall gsm network and it is also used to control the traffic load of the

19:26 BSS it must be noted that as the number of BS increases with the scaling of the subscriber population some of the

19:35 maintenance tasks are transferred to the BTS which allowing Savings in the cost of ownership of the

19:43 system the following entities have been defined operation and maintenance center that is OMC the OMC monitors and

19:53 controls all other network entities via the O interface typical OMC management functions are traffic monitoring status

20:01 20 minutes, 1 second reports of network entities subscriber and Security Management or accounting

20:08 and billing OMC has use the concept of telecommunication management Network that is tmn as standardized by the

20:16 [Music]

20:18 itu authentication center that is au as a radio interface and mobile stations are particularly vulnerable a separate

20:27 Au has been defined defined to protect user identity and data transmission the Au contains the algorithms for

20:35 authentication as well as the keys for encryption and generates the values needed for user authentication in the

20:43 hlr the Au May in fact be situated in a special protected part of the

20:50 hlr equipment identity register that is eir the eir is a database for all IM Eis

20:59 that is it stores all device identifications registered for this network as Ms are mobile they can be

21:08 easily stolen with the valid Sim anyone could use the stolen Ms the E has a

21:16 blacklist of stolen or locked devices Unfortunately The Blacklist of different providers are not usually

21:24 synchronized and the illegal use of a device in another operator network is possible the equipment identity register

21:33 also contains a list of valid imis white list and a list of malfunctioning devices that is gy

list GSM Services GSM offers much more than just voice telephony contact your local gsm network operator to the specific services that you can Avail GSM offers the three basic types of services

telepone Serv services or T services data services or beer Services supplementary Services T services the

abilities of a bearer service are used by a t service to transport data these services are further transited in the

following ways voice calls the most basic T service supported by GSM is telon this includes full rate speech at 13 kbps and emergency calls where the nearest emergency service provider is notified by dialing three digits video text and F mile another group of T

services include video text access tet text transmission F Smiley alternate speech and automatic F smiling group short text messages short messaging services SMS service is a text messaging service that allows sending and receiving text messages message on your

GSM mobile phone in addition to simple text messages other text Data including News sports Financial language and location based data can also be transmitted Bearer

Services data services or Bearer services are used through a GSM phone to receive and send data is the essential

building block leading to widespread mobile internet access and mobile data transfer

GSM currently has a data transfer rate of 9.6 kbps new development that will push up

the data transfer rates for GSM users are hscsd that is highspeed circuit switch data and GPRS that is General packet Radio Service are now available supplementary Services

supplementary services are additional services that are provided in in addition to T services and Bearer

24 minutes, 1 second Services these Services include caller identification call forwarding call waiting multi-party conversations and

barring of international outgoing calls among others a brief description of supplementary Services is given here

conferencing it allows a mobile subscriber to establish a multi-party conversation that is a simultaneous

conversation between three or more subscriber to set up a conference call this service is only applicable to

normal telephoning call waiting this service notifies a mobile

subscriber of an incoming call during a conversation the subscriber can answer

reject or ignore the incoming call

24:51 hold this service allows a subscriber to put an incoming call on hold and resume after a while the call hold service is

25:00 applicable to normal telephony call forwarding call forwarding is used to

25:07 divert calls from the original recipient to another number it is normally set up by the subscriber himself it can be used

25:16 by the subscriber to divert calls from Mobile station when the subscriber is not available call

25:24 barring call barring is useful to restrict certain types of outgoing calls such as ISD or stop incoming calls from

25:34 undesired numbers call barring is a flexible service that enables the subscriber to conditionally bar

25:42 calls number identification there are following supplementary Services related to number

25:48 identification call line identification presentation this service displays the telephone number of the calling party on

25:56 your screen calling line identification restriction a person not wishing their number to be

26:04 presented to others subscribes to this service connected line identification

26:11 presentation this service is provided to give the calling party the telephone number of the person to whom they are

26:19 connected this service is useful in situations such as forwardings where the number connected is not number

26:27 dialed connected line identification restriction there are times when the person called does not wish to have

26:34 their number presented and so they would subscribe to this person normally this overrides the presentation

26:43 service malicious call identification the malicious call identification service was provided to

26:51 combat thead of obscene or annoying calls the victim should subscribe to

26:58 this service and then they could cause unnown malicious calls to be identified

27:04 in the gsm network using a simple command advice of charge

27:11 AOC this service was designed to give the subscriber an indication of the cost of the services that they are used

27:20 furthermore those service providers who wish to offer rental services to subscribers without their own Sim can

27:28 also utilize this service in a slightly different form AOC for data calls is provided on the basis of time

27:37 measurements closed user groups cugs this service is meant for groups of

27:44 subscribers who wish to call only each other and no one else so friends this was all about GSM

27:52 system architecture hope the concepts explained in this lecture were understandable and helpful hope to see

28:00 you in the next lecture till then goodbye thank you

28:05 [Music]